



Milky Way bigger than thought

Astronomers have found evidence that the disc of the Milky Way is actually much larger than previously thought. <https://digit.in/jun18-B36-1>



Earth was a snowball

New data shows the advent of space tectonics may have transformed the early Earth into a gigantic snowball. <https://digit.in/jun18-B36-2>

you are the ruler of the world, but in most of them you don't even exist!

This is very inconvenient as a theory because a theory that allows for literally everything existing, and every outcome of every experiment, is totally non-falsifiable. We use the double negative to emphasise that all good theories in science need to not only state what evidence (when found) can prove them correct, but also what can prove them wrong. A non-falsifiable theory is just not science, and it was the biggest criticism that the No Boundary theory received.

when the inflation epoch was over that the universe began to gradually cool and matter and antimatter started popping in and out of existence. It's also because of inflation that the universe was consistent all around, or homogenous and appears the same in every direction (well almost... and it's a good thing it's not totally homogenous, or none of us would exist!).

Now, think of every universe (of which ours is just one) as a small bubble in the vast multiverse ocean – “ocean” because we can't say “space” without being totally incorrect or

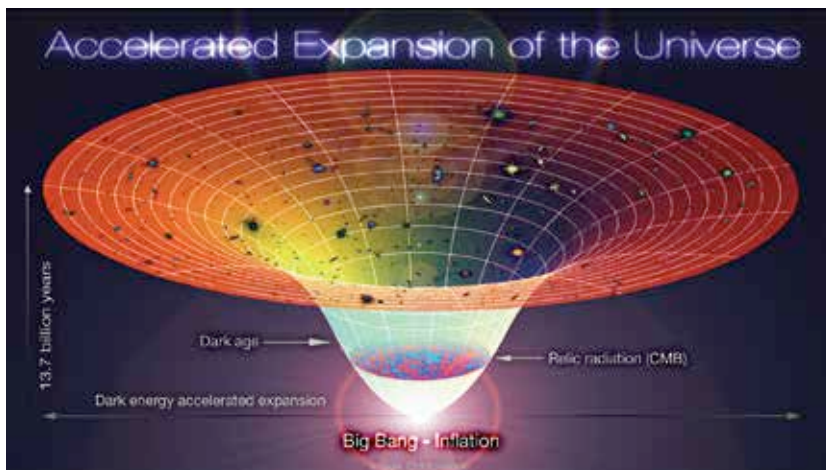
surface area of the shape that contains that space. The easiest way to explain this is to look at a black hole. Since nothing escapes the event horizon, not even light, we don't really know what's inside a black hole. The holographic principle states that the event horizon – the sphere around the black hole – contains all of the information of what's happening inside the black hole. Studying an event horizon will allow you to essentially study the black hole as well.

When extrapolated to a universe instead of a mere black hole, Hawking's latest paper actually goes against his own No Boundary theory, and suggests that the universe did indeed have an initial boundary, and its on that boundary that the imprint of the multiverse should be found – and if there is eternal inflation, that should be found as well. While not ruling out the multiverse in any way, the paper does severely limit possible universes down from infinity to a mathematically understood number.

One way to test the theory is to look for gravitational waves from the Big bang which would have the signature of eternal inflation on them. Sadly, we'd need a space based laser interferometer, which was capable of measuring much larger wavelengths of gravity waves than we can ever hope to detect with the land based LIGO, so this is not something that's happening anytime soon.

Being testable and falsifiable makes Hawking's last paper very interesting indeed. We'd learn something new if it were tested and found to be true, and we'd learn even more if it was found to be false. We're not going to know for a few decades at least. Hawking's theories are a testimony to the victory of his wondrous mind over the frail body that finally gave up on him earlier this year.

Hats off to you sir, and we look forward to your name being brought up again and again for well over the next decade. No matter what the future observations tell us, we will always have Hawking Radiation... 



The typical model of the history of our universe.

This is why Hawking worked on his new paper that essentially tried to “tame the multiverse” – something Hawking told Thomas Hertog that he wanted to do, about a year before he died. In it, they outline a way to tie together the quantum physics, which dominated the soon to be universe (just “before” the Big Bang – not really “before”, because time didn't exist yet), and the laws of physics we all accept now. They used a holographic theory that arose from string theory to tie quantum physics and classical physics together, in such a way so as to cut down the possible universes predicted by the multiverse theory to far, far less than infinity.

The No Boundary theory built on top of a theory called Eternal Inflation, where the multiverse was like a never-ending fractal design. It's only

confusing, as space is what's found inside a universe, and not a property of the multiverse.

These bubbles are formed when inflation slows or stops in a small part of the multiverse ocean, while the rest of the multiverse (or the ocean between universe bubbles) continues inflating forever. Thus, eternal inflation of the multiverse, even though inflation only happened between 10^{-36} and 10^{-32} seconds after the Big Bang in our universe.

String theory's holographic principle, (which was mostly worked out by Leonard Susskind, ironically, one of the No Boundary theory's greatest detractors), is a great fit into Hawking's and Hertog's newest paper. The holographic principle is really just a theory that all that happens in a space can be explained by looking at the